UNIVERSITY OF RWANDA
CST – School of ICT, Computer Engineering (Nyarugenge Campus)
Date: 12/11/2025

CAT-1: ITE3162 – INTERACTIVE WEB DEVELOPMENT – 30marks in 1 hour 30 minutes

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QUESTION 1

a. Python is known for its readability and flexibility. Discuss the key language features that contribute to its popularity among developers and analyze how its interpreted nature affects performance compared to compiled languages like C or Java. **/3 marks**

b. You are given two lists, one containing medication attributes and the other containing their corresponding details. Write a Python function that creates a dictionary from the two lists. (keys = ['drug_name', 'dosage', 'category'], values = ['Paracetamol', '500mg', 'Pain Reliever']). **/4 marks**

c. Write a Python program that prompts the user to enter two consecutive ages of children. Your program should validate the input to ensure both are integers, then display their sum and difference. **/3 marks**

QUESTION 2

a. Briefly define method overriding and encapsulation in Python. **/2 marks**
b. Student Grade Evaluator: Input a student's marks in five subjects, calculate the average, and assign a grade using if-elif-else. Store marks in a list and use a function to compute the average. **/4 marks**
c. Build a program where users can register with a username and password and then log in. Use file writing for registration and file reading for login validation. Include clear success/error messages. **/4 marks**

QUESTION 3

a. Given a class LibraryMember that includes attributes for member_name, borrowed_books, and membership_level. Implement methods for borrowing and returning books, ensuring that members cannot borrow more books than their membership level allows. Include a method to display the member's current borrowed books. **/5 marks**

b. Create a Python package named farm_tools containing two modules: crop_math.py (for calculating total yield and average yield per hectare) and land_geometry.py (for computing the area of rectangular and circular farm plots). Write a main program that imports and uses these modules to perform basic farm-related calculations, ensuring a proper package structure and imports. **/5 marks**

CamScanner